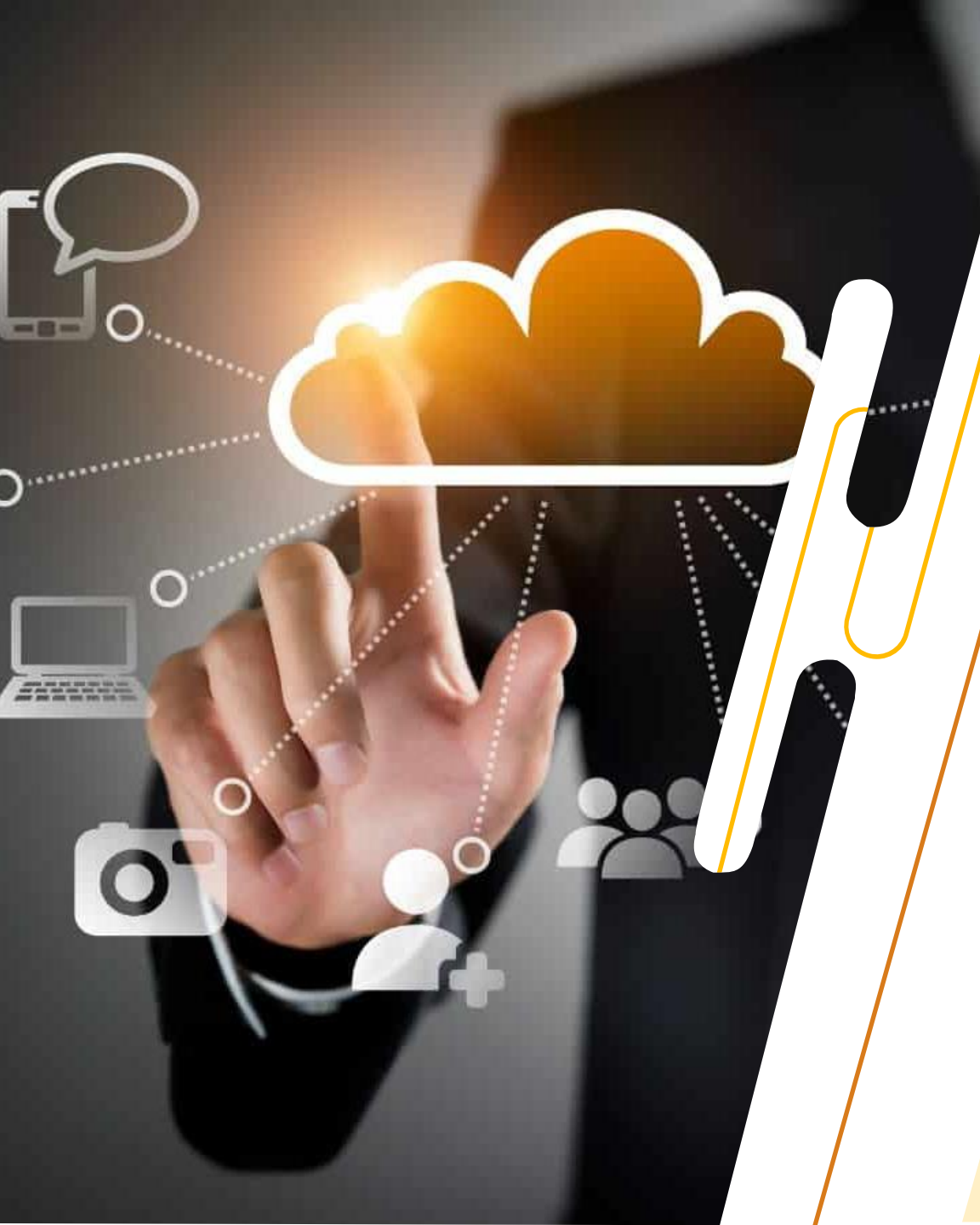




IT3061

Massive Data Processing and Cloud Computing

YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

Lecture Outline

- 1. What is Cloud Computing?
- 2. Evolution of Cloud Computing
- 3. Definition of Cloud Computing
- 4. Characteristics of Cloud Computing
- 5. Cloud Service Models
- 6. Cloud Deployment Models
- 7. Who are Cloud Service Providers?





WHAT IS A DATA CENTER?



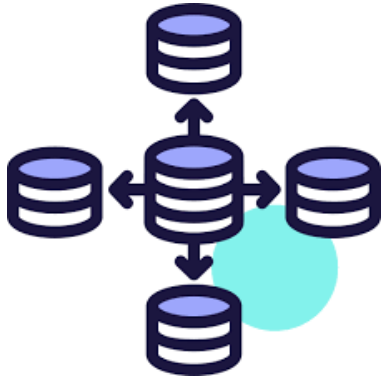
Processing



Data Center



Collecting



Distributing



Allow access



Storing

amazon

Google

Microsoft

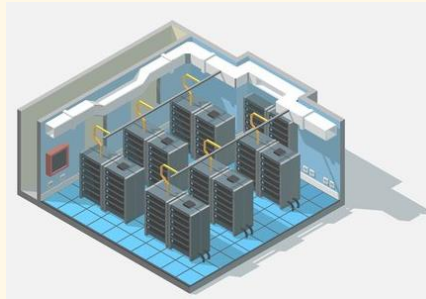


Challenges of managing a data center

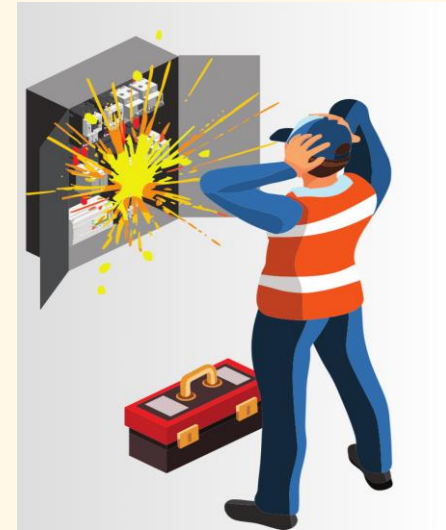
Cost



Space



Maintenance
issues



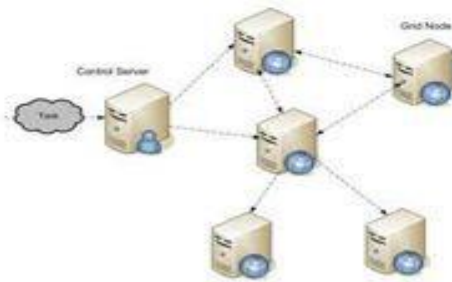


Cloud Computing

The Evolution of Cloud Computing

Grid Computing

- Solving large problems with parallel computing
- Made mainstream by Globus Alliance



Utility Computing

- Offering computing resources as a metered service
- Introduced in late 1990s



Software as a Service

- Network-based subscription to applications
- Gained momentum in 2001



Cloud Computing

- Next-Generation Internet computing
- Next Generation Data Centers



The NIST (National Institute of Standards and Technologies) Definition of Cloud Computing

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.



Characteristics of Cloud Computing

Resource Pooling

share resources among several clients, providing everyone with a different set of services as per their requirements.

On-Demand & Self-Service

enables the client to constantly monitor the server uptime, abilities, and allotted network storage.

Elasticity & Scalability

Elasticity implies that manufacturing organizations can rapidly provision and de-provision any of the cloud computing resources.

Characteristics of Cloud Computing

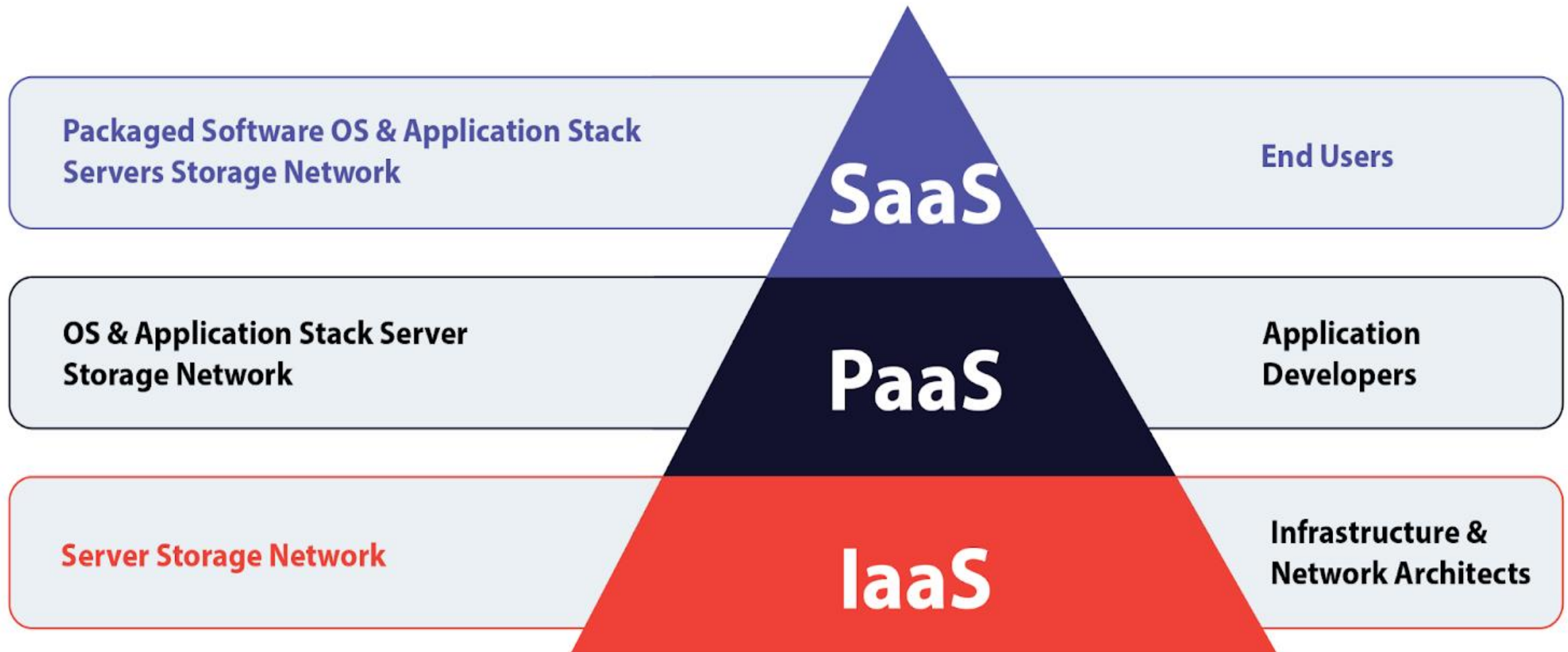
Measured Service

Cloud computing resources usage is metered and manufacturing organizations pay accordingly for what they have used

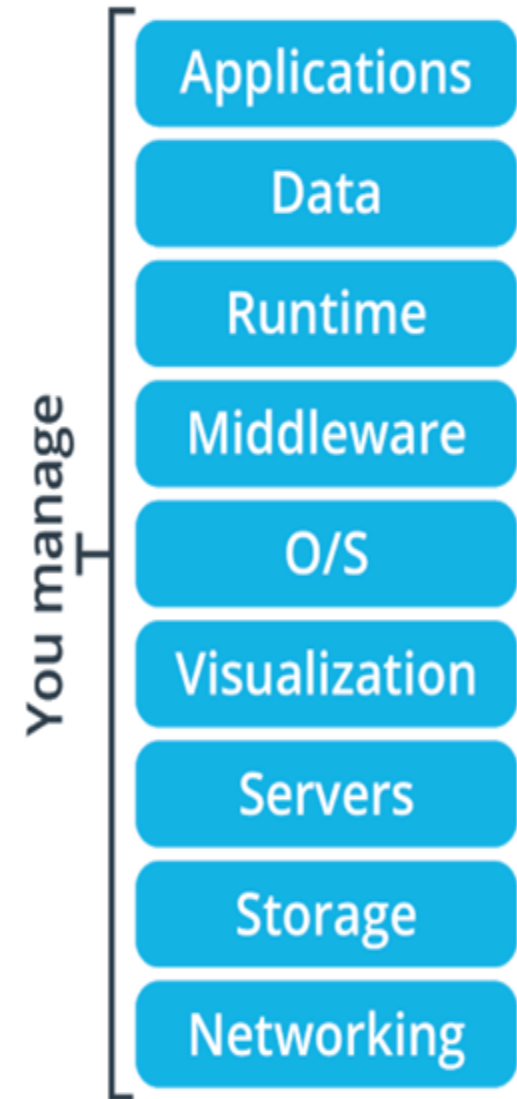
Broad Access Network

Cloud computing resources are available over the network and can be accessed by diverse customer platforms

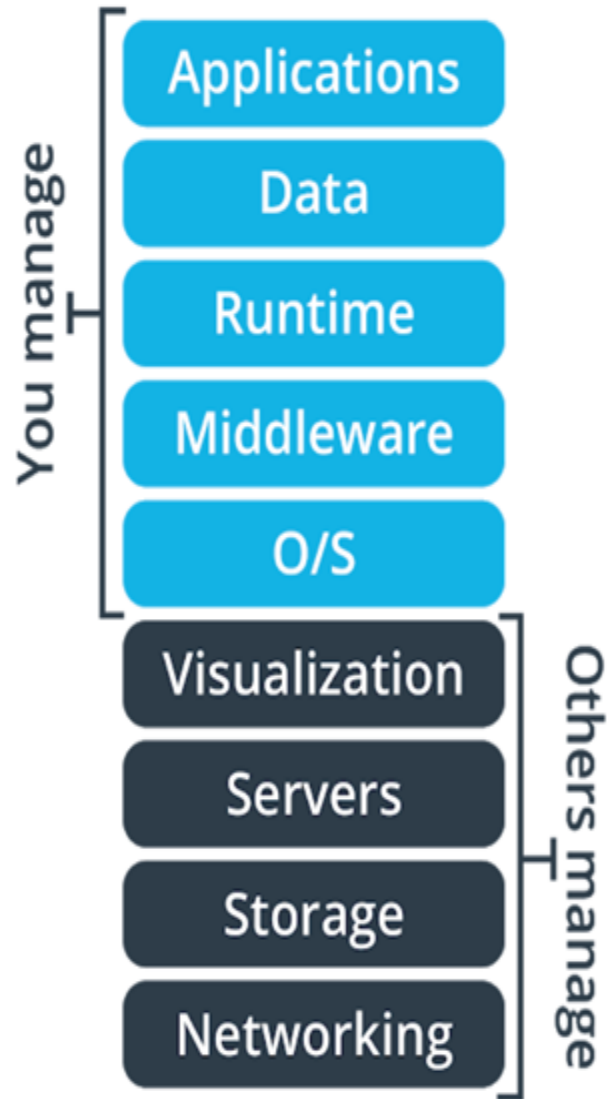
Cloud Service Models



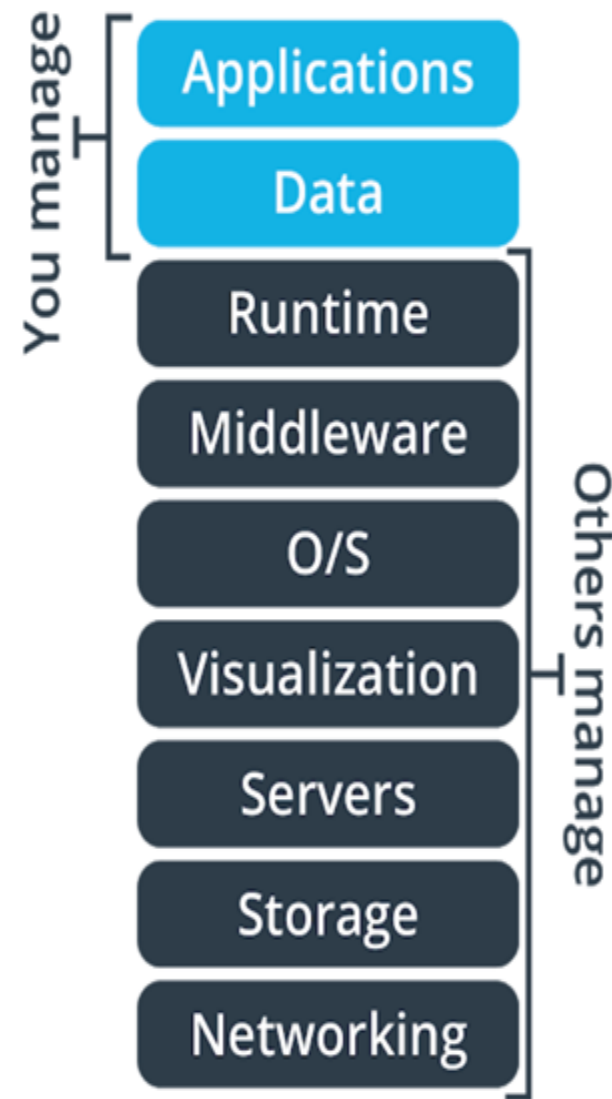
On-Premises



IaaS (Infrastructure as a Service)



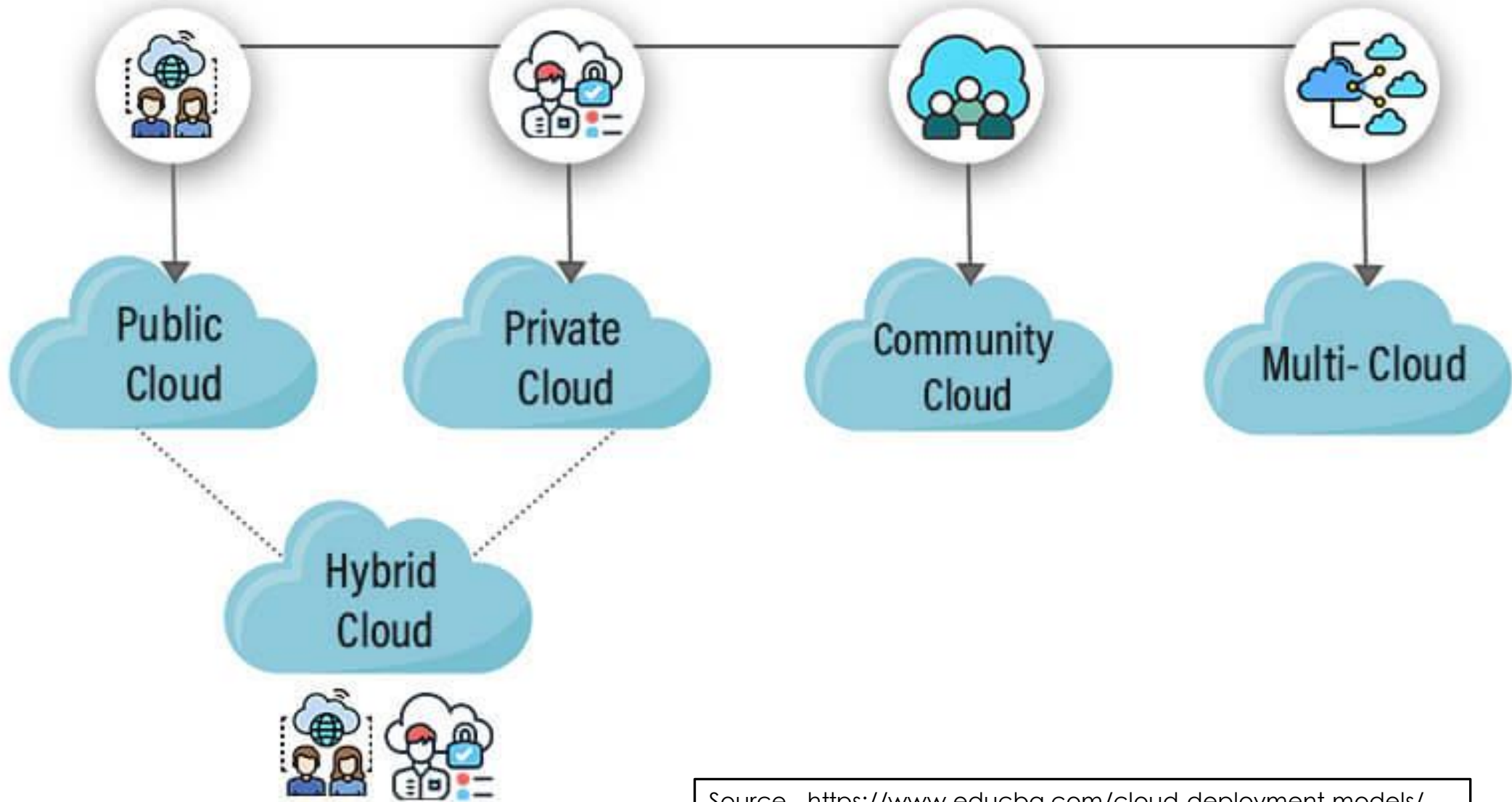
PaaS (Platform as a Service)



SaaS (Software as a Service)



Cloud Deployment Models



Source - <https://www.educba.com/cloud-deployment-models/>



Types of Cloud Deployment

Public Cloud

Typically have massive amounts of available space, which translates into easy scalability. Recommended for software development and collaborative projects.

Private Cloud

Usually reside behind a firewall and are utilized by a single organization. Recommended for businesses with very tight regulatory requirements

Hybrid Cloud

Combine public clouds with private clouds to allow the two platforms to interact seamlessly. Recommended for businesses balancing big data analytics with strict data privacy regulations.

Community Cloud

A collaborative, multi-tenant platform used by several distinct organizations to share the same applications. Users are typically operating within the same industry or field.

Cloud Service Models

IaaS

Infrastructure As A Service

PaaS

Platform As A Service

SaaS

Software As A Service

Private
Cloud

Community
Cloud

Public
Cloud

Hybrid
Cloud

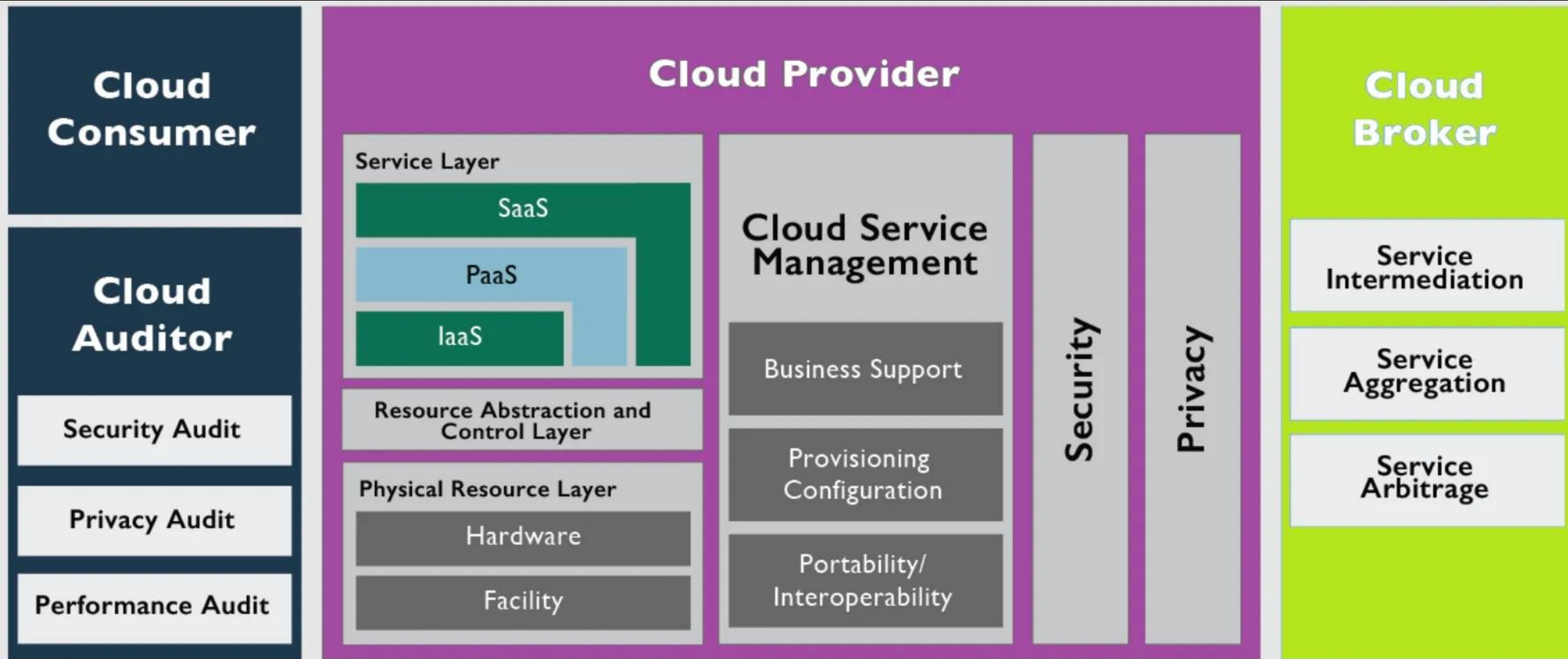
Cloud Deployment Models

Who are Cloud Service Providers (CSP)?

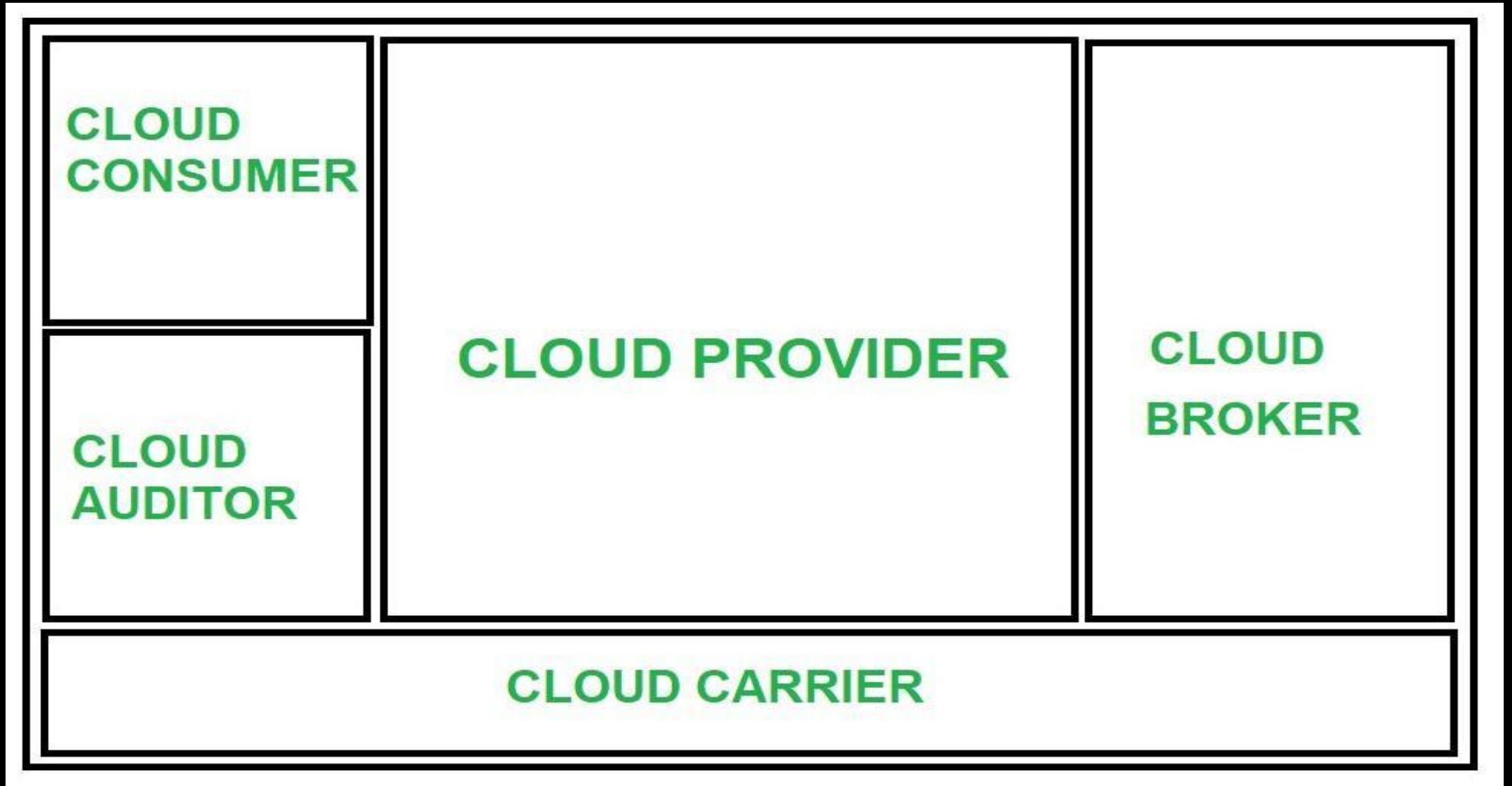
A cloud service provider is a third-party company offering a cloud-based platform, infrastructure, application, or storage services.



NIST Cloud Computing Architecture



NIST Cloud Computing Architecture

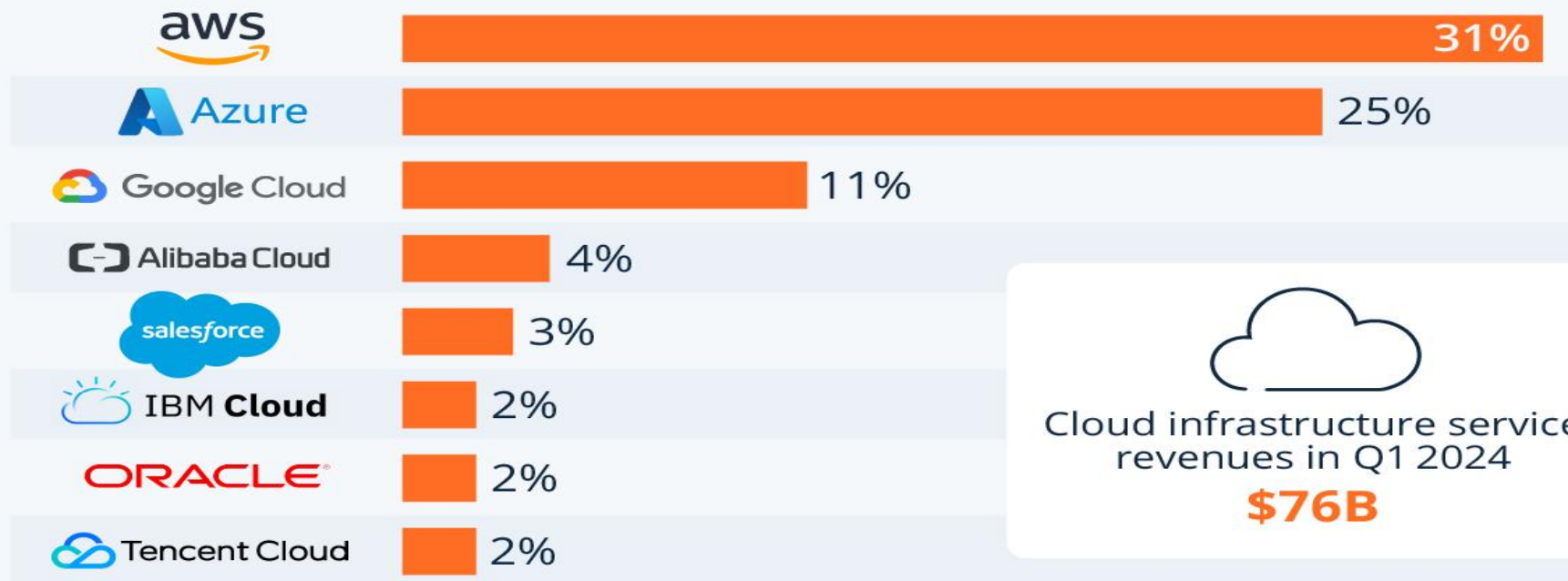


Top 10 Cloud Providers



Amazon Maintains Cloud Lead as Microsoft Edges Closer

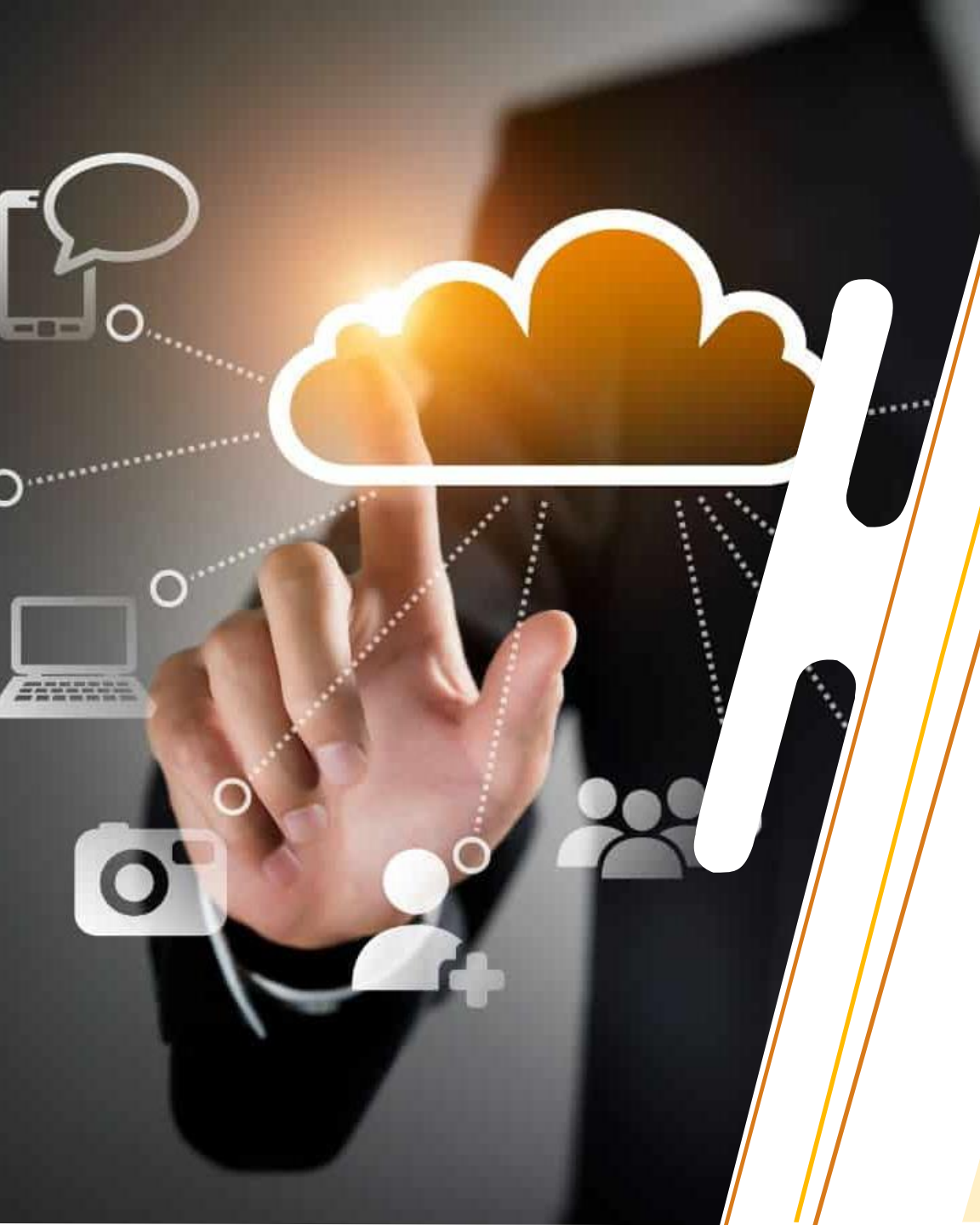
Worldwide market share of leading cloud infrastructure service providers in Q1 2024*



* Includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group





Next Session :
Cloud Core
Services